

ESKF State-Transition Jacobian: Continuous Error-State and Direct Discrete Derivations

1 Purpose

This note connects two ways of deriving the ESKF state-transition matrix F :

1. derive the continuous-time error-state kinematics and then discretize them;
2. start directly from the discrete nominal-state propagation and linearize it with respect to local state perturbations.

They describe the same propagation of small errors. When the same approximation order is retained, the results agree. Direct differentiation of a discrete propagation may retain higher-order terms in Δt that are absent from a first-order discretization $F_d \approx I + F_c \Delta t$.

2 True, nominal, and error states

The true state is represented by a nominal state plus a small error. For Euclidean variables,

$$\mathbf{p}^t = \mathbf{p} + \delta \mathbf{p}, \tag{1}$$

$$\mathbf{v}^t = \mathbf{v} + \delta \mathbf{v}, \tag{2}$$

$$\mathbf{b}_g^t = \mathbf{b}_g + \delta \mathbf{b}_g, \tag{3}$$

$$\mathbf{b}_a^t = \mathbf{b}_a + \delta \mathbf{b}_a, \tag{4}$$

$$\mathbf{g}^t = \mathbf{g} + \delta \mathbf{g}. \tag{5}$$

For rotation, using a right perturbation,

$$\mathbf{R}^t = \mathbf{R} \text{Exp}(\delta \boldsymbol{\theta}). \tag{6}$$

The 18-dimensional error state is

$$\delta \mathbf{x} = [\delta \mathbf{p} \quad \delta \mathbf{v} \quad \delta \boldsymbol{\theta} \quad \delta \mathbf{b}_g \quad \delta \mathbf{b}_a \quad \delta \mathbf{g}]^T. \tag{7}$$

3 Two routes to the discrete transition matrix

The textbook route starts from continuous-time error dynamics,

$$\delta \dot{\mathbf{x}} = F_c \delta \mathbf{x} + G_c \mathbf{w}, \tag{8}$$

and then discretizes them. A first-order discretization gives

$$F_d \approx I + F_c \Delta t. \tag{9}$$

The direct discrete route first writes the nominal propagation

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k), \quad (10)$$

perturbs the input state, propagates it through the same discrete function, and measures the resulting output perturbation:

$$\mathbf{x}_k^t = \mathbf{x}_k \oplus \delta \mathbf{x}_k, \quad (11)$$

$$\mathbf{x}_{k+1}^t = f_d(\mathbf{x}_k \oplus \delta \mathbf{x}_k, \mathbf{u}_k), \quad (12)$$

$$\mathbf{x}_{k+1}^t = \mathbf{x}_{k+1} \oplus \delta \mathbf{x}_{k+1}. \quad (13)$$

Linearization gives

$$\delta \mathbf{x}_{k+1} \approx F_d \delta \mathbf{x}_k. \quad (14)$$

Thus F_d is the local Jacobian describing how the current error propagates into the next error.

4 Discrete nominal-state propagation

Using the Chapter 3 notation, the IMU measurements are denoted by $\tilde{\mathbf{a}}$ and $\tilde{\boldsymbol{\omega}}$. The increments used in propagation are

$$\boldsymbol{\tau} \triangleq (\tilde{\boldsymbol{\omega}} - \mathbf{b}_g) \Delta t. \quad (15)$$

The nominal propagation considered here is

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a) \Delta t^2 + \frac{1}{2} \mathbf{g}_k \Delta t^2, \quad (16)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a) \Delta t + \mathbf{g}_k \Delta t, \quad (17)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \text{Exp}(\boldsymbol{\tau}), \quad (18)$$

$$\mathbf{b}_{gk+1} = \mathbf{b}_{gk}, \quad (19)$$

$$\mathbf{b}_{ak+1} = \mathbf{b}_{ak}, \quad (20)$$

$$\mathbf{g}_{k+1} = \mathbf{g}_k. \quad (21)$$

5 Position row of F

Starting from

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a) \Delta t^2 + \frac{1}{2} \mathbf{g}_k \Delta t^2, \quad (22)$$

we obtain immediately

$$F_{pp} = I, \quad (23)$$

$$F_{pv} = I \Delta t. \quad (24)$$

For the rotation term, apply a right perturbation:

$$\mathbf{R}_k^t = \mathbf{R}_k \text{Exp}(\delta \boldsymbol{\theta}_k) \approx \mathbf{R}_k (I + \delta \boldsymbol{\theta}_k^\wedge). \quad (25)$$

Then

$$\mathbf{R}_k^t (\tilde{\mathbf{a}} - \mathbf{b}_a) \approx \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a) + \mathbf{R}_k \delta \boldsymbol{\theta}_k^\wedge (\tilde{\mathbf{a}} - \mathbf{b}_a) \quad (26)$$

$$= \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a) - \mathbf{R}_k (\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge \delta \boldsymbol{\theta}_k, \quad (27)$$

where $\delta\theta^\wedge(\tilde{\mathbf{a}} - \mathbf{b}_a) = -(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge\delta\theta$. Therefore

$$\boxed{F_{p\theta} = -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge\Delta t^2.} \quad (28)$$

Because the accelerometer term is $\tilde{\mathbf{a}} - \mathbf{b}_a$,

$$\boxed{F_{pb_a} = -\frac{1}{2}\mathbf{R}_k\Delta t^2.} \quad (29)$$

The gravity contribution is

$$\boxed{F_{pg} = \frac{1}{2}I\Delta t^2.} \quad (30)$$

There is no direct b_g dependence in this discrete position update, hence

$$F_{pb_g} = 0. \quad (31)$$

The complete position error propagation is therefore

$$\delta\mathbf{p}_{k+1} = \delta\mathbf{p}_k + \Delta t \delta\mathbf{v}_k - \frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge\Delta t^2\delta\theta_k \quad (32)$$

$$- \frac{1}{2}\mathbf{R}_k\Delta t^2\delta\mathbf{b}_{ak} + \frac{1}{2}I\Delta t^2\delta\mathbf{g}_k. \quad (33)$$

6 Velocity row of F

From

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)\Delta t + \mathbf{g}_k\Delta t, \quad (34)$$

we obtain

$$F_{vp} = 0, \quad (35)$$

$$F_{vv} = I, \quad (36)$$

$$\boxed{F_{v\theta}} = \boxed{-\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge\Delta t}, \quad (37)$$

$$F_{vb_g} = 0, \quad (38)$$

$$\boxed{F_{vb_a}} = \boxed{-\mathbf{R}_k\Delta t}, \quad (39)$$

$$\boxed{F_{vg}} = \boxed{I\Delta t}. \quad (40)$$

Thus

$$\delta\mathbf{v}_{k+1} = \delta\mathbf{v}_k - \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge\Delta t\delta\theta_k - \mathbf{R}_k\Delta t\delta\mathbf{b}_{ak} + I\Delta t\delta\mathbf{g}_k. \quad (41)$$

7 Rotation row of F and the manif Jacobians

The rotation propagation is

$$\mathbf{R}_{k+1} = \mathbf{R}_k \oplus \boldsymbol{\tau} = \mathbf{R}_k \text{Exp}(\boldsymbol{\tau}), \quad \boldsymbol{\tau} = (\tilde{\boldsymbol{\omega}} - \mathbf{b}_g)\Delta t. \quad (42)$$

A `manif` call of the form

`new_R = R_.rplus(tau, J_R, J_tau);`

returns the propagated rotation together with the local Jacobians

$$J_R = \frac{\partial(\mathbf{R} \oplus \boldsymbol{\tau})}{\partial \mathbf{R}}, \quad (43)$$

$$J_\tau = \frac{\partial(\mathbf{R} \oplus \boldsymbol{\tau})}{\partial \boldsymbol{\tau}}. \quad (44)$$

For the right-plus convention on $SO(3)$,

$$\boxed{J_R = \text{Exp}(-\boldsymbol{\tau})}, \quad (45)$$

$$\boxed{J_\tau = J_r(\boldsymbol{\tau})}. \quad (46)$$

Therefore

$$\boxed{F_{\theta\theta} = J_R = \text{Exp}(-\boldsymbol{\tau})}. \quad (47)$$

Since

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathbf{b}_g} = -I \Delta t, \quad (48)$$

the chain rule gives

$$\boxed{F_{\theta b_g} = -J_\tau \Delta t = -J_r(\boldsymbol{\tau}) \Delta t}. \quad (49)$$

Hence

$$\delta \boldsymbol{\theta}_{k+1} = J_R \delta \boldsymbol{\theta}_k - J_\tau \Delta t \delta \mathbf{b}_{gk}. \quad (50)$$

For a small IMU interval,

$$J_R = \text{Exp}(-\boldsymbol{\tau}) \approx I - (\tilde{\boldsymbol{\omega}} - \mathbf{b}_g)^\wedge \Delta t + O(\Delta t^2), \quad (51)$$

$$J_r(\boldsymbol{\tau}) \approx I - \frac{1}{2} \boldsymbol{\tau}^\wedge + O(\|\boldsymbol{\tau}\|^2). \quad (52)$$

Consequently

$$-J_r(\boldsymbol{\tau}) \Delta t = -I \Delta t + O(\Delta t^2), \quad (53)$$

which explains the textbook first-order approximation $F_{\theta b_g} \approx -I \Delta t$.

8 Bias and gravity rows

With random-walk driving noise omitted from the deterministic state-transition Jacobian,

$$\delta \mathbf{b}_{gk+1} = \delta \mathbf{b}_{gk}, \quad (54)$$

$$\delta \mathbf{b}_{ak+1} = \delta \mathbf{b}_{ak}, \quad (55)$$

$$\delta \mathbf{g}_{k+1} = \delta \mathbf{g}_k. \quad (56)$$

Thus

$$F_{b_g b_g} = I, \quad F_{b_a b_a} = I, \quad F_{g g} = I, \quad (57)$$

and their other deterministic blocks are zero.

9 Complete direct-discrete state-transition matrix

Using

$$\boldsymbol{\tau} = (\tilde{\boldsymbol{\omega}} - \mathbf{b}_g)\Delta t, \quad (58)$$

the direct Jacobian of the stated discrete nominal propagation is

$$F_d = \begin{bmatrix} I & I\Delta t & -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge \Delta t^2 & 0 & -\frac{1}{2}\mathbf{R}_k\Delta t^2 & \frac{1}{2}I\Delta t^2 \\ 0 & I & -\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge \Delta t & 0 & -\mathbf{R}_k\Delta t & I\Delta t \\ 0 & 0 & J_R & -J_\tau\Delta t & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & 0 & I \end{bmatrix}, \quad (59)$$

where

$$J_R = \text{Exp}(-\boldsymbol{\tau}), \quad J_\tau = J_r(\boldsymbol{\tau}). \quad (60)$$

10 Why the textbook position row is simpler

For the continuous error-state system, the position equation is

$$\delta \dot{\mathbf{p}} = \delta \mathbf{v}. \quad (61)$$

Therefore the continuous-time block from rotation error to position-error rate is zero:

$$(F_c)_{p\theta} = 0. \quad (62)$$

If the discrete transition is approximated only to first order,

$$F_d \approx I + F_c\Delta t, \quad (63)$$

then

$$(F_d)_{p\theta} = 0. \quad (64)$$

The direct discrete position equation, however, contains the term

$$\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)\Delta t^2, \quad (65)$$

so its Jacobian contains

$$F_{p\theta} = -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_a)^\wedge \Delta t^2. \quad (66)$$

This is a second-order term. It is not a contradiction: it is simply discarded by the first-order approximation $I + F_c\Delta t$. A higher-order expansion

$$e^{F_c\Delta t} = I + F_c\Delta t + \frac{1}{2}F_c^2\Delta t^2 + \dots \quad (67)$$

recovers such indirect couplings, e.g. rotation error $\rightarrow$ velocity error $\rightarrow$ position error.

Similarly,

$$J_R = \text{Exp}(-(\tilde{\boldsymbol{\omega}} - \mathbf{b}_g)\Delta t) = I - (\tilde{\boldsymbol{\omega}} - \mathbf{b}_g)^\wedge \Delta t + O(\Delta t^2), \quad (68)$$

so the exact/local discrete Lie-group Jacobian and the textbook first-order rotation block agree to first order.

11 Main interpretation

The essential quantity is not “the nominal-state Jacobian” versus “the error-state Jacobian” as two unrelated objects. The transition matrix F_d describes how a small state error before propagation becomes a small state error after propagation:

$$\boxed{\delta \mathbf{x}_{k+1} \approx F_d \delta \mathbf{x}_k.} \tag{69}$$

There are two useful derivation routes:

continuous route: true kinematics $\rightarrow$ continuous error dynamics $\rightarrow F_c \rightarrow F_d$,
discrete route: discrete nominal propagation $\rightarrow$ local perturbation $\rightarrow F_d$.

The **manif** Jacobians implement the second viewpoint naturally for Lie-group variables such as $R \in SO(3)$. For Euclidean variables such as p and v , the same idea reduces to ordinary differentiation.

12 Implementation consistency note

If the discrete equations for p_{k+1} and v_{k+1} use the pre-propagation rotation R_k , their direct Jacobians should also be evaluated at R_k . Updating the stored rotation to R_{k+1} before constructing those Jacobian blocks changes the linearization point and is not exactly the Jacobian of the discrete equations written above.